1.1 What is Big Data

to Big Data

Definition:

- Big Data is often described as extremely large data sets that have grown beyond the ability to manage and analyze them with traditional data processing tools. The data set has grown so large that it is difficult to manage and even harder to garner value out of it.
- The primary difficulties are the acquisition, storage, searching, sharing, analytics, and visualization of data. Not only the size of the data set but also difficult to process the data

The data come from everywhere: Sensors used to gather climate information, posts to social media sites, digital pictures and videos posted online, transaction records of online purchases, and cell phone GPS signals etc. All of these data have intrinsic value that can be extracted using analytics, algorithms, and other technique

Characteristics of Big Data

Three characteristics define Big Data: ***volume, variety,*** **and** ***velocity***

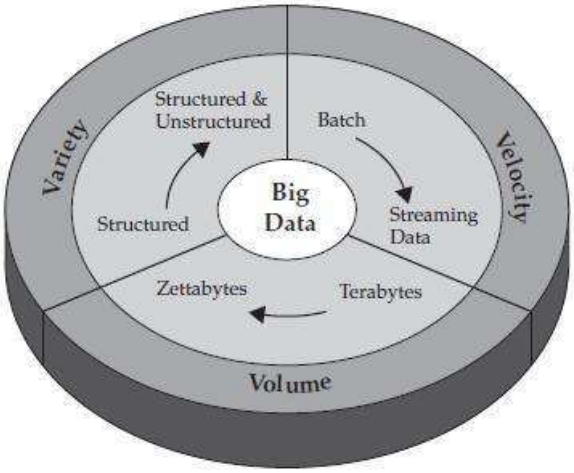

Fig1: Big Data Characteristics

Volume: The amount of data

The size of available data has been growing at an increasing rate. The volume of data is growing. Experts predict that the volume of data in the world will grow to 35 Zetta bytes in 2020.

- Twitter alone generates more than 7 terabytes (TB) of data every day. Facebook 10TB
- That same phenomenon affects every business – their data is growing at the same exponential rate too.
- This applies to companies and to individuals. A text file is a few kilo bytes, a sound file is a few meg bytes while a full-length movie is a few giga bytes. More sources of data are added on continuous basis.
- For companies, in the old days, all data was generated internally by employees. Currently, the data is generated by employees, partners and customers.
- For a group of companies, the data is also generated by machines. For example, Hundreds millions of smart phones send a variety of information to the network infrastructure.
- We store everything: Environmental data, financial data, medical data, surveillance data. Petabyte datasets are common these days and Exabyte is not far away.

Velocity: How fast it is generated

- Data is increasingly accelerating the velocity at which it is created and at which it is integrated We have moved from batch to a real-time business.
- Initially, companies analyzed data using a batch process. One takes a chunk of data, submits a job to the server and waits for delivery of the result. That scheme works when the incoming data rate is slower than the batch -processing rate and when the result is useful despite the delay.
- With the new sources of data such as social and mobile applications, the batch process breaks down. The data is now streaming into the server in real time, in a continuous fashion and the result is only useful if the delay is very short.

Variety: Represents all kinds of data

Data can be classified under several categories: structured data, semi structured data and unstructured data.

Structured data are normally found in traditional databases (SQL or others) where data are organized into tables based on defined business rules. Structured data usually prove to be the easiest type of data to work with, simply because the data are defined and indexed, making access and filtering easier.

Unstructured data, are not organized into tables and cannot be natively used by applications or interpreted by a database. A good example of unstructured data would be a collection of binary image files.

Semi structured data fall between unstructured and structured data. Semi structured data do not have a formal structure like a database with tables and relationships. However, unlike unstructured data, semi structured data have tags or other markers to separate the elements and provide a hierarchy of records and fields, which define the data.

- Big data extend beyond structured data to include unstructured data off all varieties: text, audio, video, click streams, log files and more.
- The growth in data sources has fueled the growth in data types. In fact, 80% of the world's data is unstructured and only 20% structured data Yet most traditional methods apply analytics only to structured information.

1.2 Why Big Data is Important

- Big Data solutions are ideal for analyzing not only raw structured data, but semi structured and unstructured data from a wide variety of sources.
- Big Data solutions are ideal when all, or most, of the data needs to be analyzed versus a sample of the data; or a sampling of data isn't nearly as effective as a larger set of data from which to derive analysis.
- Big Data solutions are ideal for iterative and exploratory analysis when business measures on data are not predetermined.

- Big Data is well suited for solving information challenges that don't natively fit within a traditional relational database approach for handling the problem at hand.
- Big Data has already proved its importance and value in several areas. Organizations such as the National Oceanic and Atmospheric Administration (NOAA), the National Aeronautics and Space Administration (NASA), several pharmaceutical companies, and numerous energy companies have amassed huge amounts of data and now leverage Big Data technologies on a daily basis to extract value from them.

  NOAA uses Big Data approaches to aid in climate, ecosystem, weather, and commercial research,
- NASA uses Big Data for aeronautical and another research.
- Pharmaceutical companies and energy companies have leveraged Big Data for more tangible results. such as drug testing and geophysical analysis.
- The New York Times has used Big Data tools for text analysis and Web mining
- Walt Disney Company uses them to correlate and understand customer behavior in all of its stores, theme parks.
- Companies such as Facebook, Amazon, and Google rely on Big Data analytics a part of their primary marketing schemes as well as a means of servicing their customers better.
- This accomplished by storing each customer's searches and purchases and other piece of information av aimable, and then applying algorithms to that information to compare one customer's information with all other customers information.
- Big Data plays another role in today's businesses: Large organizations increasingly face the need to maintain massive amounts of structured and unstructured data — from transaction information in data warehouses to employee tweets, from supplier records to regulatory filings—to comply with government regulations.